Short-mode white noise in a radiation universe: spatial curvature and the measured temperature

Opus 5.5, Meatbag Operator

2026-11-04

Verdict

Two short sound waves of nearly equal wavelength beat, and their quadratic stress has a slow component at the beat wavenumber K. It builds one trace-free tensor, $S_{ij} \to W(x)\,\mathrm{TF}(\hat{q}\hat{q})$. The volume variable ζ cannot see it, the curvature $\mathcal{R}$ reads it along $\hat{K}$, and the sky reads its change in time along $\hat{n}$.

Results are at leading order in amplitude, $O(\epsilon^2)$ in the fields and $O(\epsilon^4)$ in the spectra, with $K \to 0$ at fixed finite time and $\epsilon q t \ll 1$. The arguments are analytic; numerical corroboration is in Appendix A.

Spatial curvature: yes. Short-wavelength modes alone give the specified $\mathcal{R}$ a nonzero white-noise coefficient, $P_{\mathcal{R}}(K) \to \epsilon^4 A_0(t) > 0$, with no $1/K^2$ enhancement and no suppression by powers of K. The coefficient is explicit, Eqs. (7)–(9). Its direction-dependent part is the projection of S_{ij}, and it grows as $\ln(qt)$ once the short modes are inside the horizon. The answer depends on the threading in the definition of $\mathcal{R}$: the conserved ζ has no response at all, to all orders.

Measured temperature: yes, but not as curvature. The complete normalised map has a nonzero K^0 quadrupole. By Theorem 1, $\mathcal{Q}_2 = -\frac{1}{3}[W]^o_\star + \mathcal{Q}_2^{\rm em} + B$ with B bounded, so $\mathcal{Q}_2 = -8\ln(t_o/t_\star) + \mathcal{Q}_2^{\rm em} + O(1)$: the sky sees the time *difference* of the tensor in $\mathcal{R}$. White noise alone is not the signal, since a finite $K \to 0$ limit is generic for observer-local quantities (Prop. 2). The direct emission term has zero mean but rms growing as $qt_\star$, so it dominates the power once $qt_\star \gg \ln(t_o/t_\star)$.

This is the first of the three outcomes: an explicit nonzero coefficient. Not established: the exchange of $K \to 0$ with $\epsilon \to 0$, and a complete C_ℓ, which a white-noise $P(K)$ does not determine.

1 Setup and the definition of $\mathcal{R}$

Conventions and the field equations (N), (T), (M), (Ham) are those of the main report. For an irrotational radiation fluid, $Tu_\mu = -\mathcal{T}\partial_\mu t$ defines a potential t, and the X^i are comoving material labels. In these coordinates

$$g = N^2(-\mathrm{d}t^2 + \gamma_{ij}\mathrm{d}X^i\mathrm{d}X^j), \qquad N = a_1 t e^\nu = \mathcal{T}/T, \qquad \gamma = e^{2\zeta}e^H, \quad \mathrm{tr}\,H = 0. \tag{1}$$

Here ζ is the volume of γ, H its shape, and ν the local clock. The primordial state is the regular solution with $\zeta = \epsilon\zeta_1(X)$ exactly and ζ_1 Gaussian, with $\gamma \to e^{2\zeta}\delta$ and $\nu, H \to 0$ as $t \to 0$. The spectrum is smooth with support in $q_{\min} < q < q_{\max}$.

Definition of $\mathcal{R}$. *Slicing:* level sets of t, the hypersurfaces orthogonal to u. *Threading:* on each slice, the chart x in which the induced metric takes the form

$$h_{ij} = a^2 e^{2\mathcal{R}}(e^\chi)_{ij}, \qquad \mathrm{tr}\,\chi = 0, \quad \partial_j\chi_{ij} = 0, \qquad a = a_1 t. \tag{2}$$

Geometrically, $\mathcal{R}$ is the conformal factor of the comoving slice in the chart where the unimodular part is transverse, so χ does not enter the scalar curvature linearly and $a^2\,{}^{(3)}R = -4\nabla^2\mathcal{R} + 8\mathcal{R}\nabla^2\mathcal{R} - 2(\partial\mathcal{R})^2 + \ldots$. It is a nonlocal functional of the 3-geometry; the chart exists perturbatively and is unique at nonzero wavenumber. With $X = x + \xi(x,t)$ and $J = \det \partial X/\partial x$, taking the determinant gives the exact relation

$$\mathcal{R}(x,t) = \zeta(X) + \nu(X,t) + \tfrac{1}{3}\ln J. \tag{3}$$

As $t \to 0$, $\mathcal{R} = \zeta$ exactly. Through second order, with $\xi_1 = \partial\beta_1$, $m = \partial\partial\beta_1$, $\psi = \zeta + \nu$, and first-order threading fixed by $\chi_1 = 0$,

$$\mathcal{R}_2 = \psi_2 + \xi_1 \cdot \partial\psi_1 - \tfrac{1}{6}\operatorname{tr} m^2 - \tfrac{1}{4}\hat{K}_i\hat{K}_j S_{ij}, \qquad S = H_2 + \xi_1 \cdot \partial H_1 - \mathrm{TF}\, m^2. \tag{4}$$

At first order, $x = qt/\sqrt{3}$, $\nu_1 = \cos x - j_0(x)$, $j_0 = \sin x/x$, and $H_1 = h(x)\,\mathrm{TF}(\hat{q}\hat{q})\zeta_1$ with $h = 6(1+\cos x - 2j_0)$. This gives $\mathcal{R}_1 = j_0(x)\zeta_1$, which starts at 1 and decays inside the horizon.

2 First route: the decisive calculation

Take the seed $\zeta_1 = A\cos(q\cdot x) + B\cos[(q-K)\cdot x]$. The output at K is $(AB/2)\cos(K\cdot x)\,K_2$, where $K_2 = 2F_2$ is the coefficient of $z_1 z_2$ for the pair $(q, K-q)$ with $z_1 z_2 = AB/4$.

Why the limit is finite and direction dependent. The trace-free equation at second order is

$$H_2'' + \tfrac{2}{t}H_2' + \mathfrak{B}_K H_2 = -4\,\mathrm{TF}(K\otimes K)\nu_2 - 2c_T, \qquad \mathfrak{B}_K A = \mathrm{TF}[K^2 A - K\otimes(AK) - (AK)\otimes K],$$

with c_T built from first-order fields. Its regular Green function $\frac{s}{t}\sum_n (-1)^n (t-s)^{2n+1}\mathfrak{B}_K^n/(2n+1)!$ is entire in the Cartesian components of K, with no projector and no inverse power of $|K|$. So is the regular solution of the lapse equation $3\nu_2'' + K^2\nu_2 - 6\nu_2/t^2 + c_N = 0$ (main report, "Projector-free Cartesian solution"). Hence H_2 and ν_2 are continuous at $K = 0$, with $H_2 \to a(x)\,\mathrm{TF}(\hat{q}\hat{q})$ and $\nu_2 \to n_2(x)$, and no pole can arise from them. But the projector in (4) remains, and $\hat{K}\hat{K}\cdot\mathrm{TF}(\hat{q}\hat{q}) = \mu^2 - \frac{1}{3}$, with $\mu = \hat{K}\cdot\hat{q}$. So a constant survives, and it is not analytic in K.

The $K = 0$ pair fields. With ζ along one axis the equations reduce to 1+1 dimensions, with sources from first-order fields. With $q = 1$, (Ham) at $K = 0$ is first order and (T) is second order:

$$(xn_2)_x = 6x\sin x + 2x\sin 2x + 5\cos 2x - \frac{5\sin 2x}{x} + \frac{5\sin^2 x}{x^2}, \qquad a'' + \frac{2}{t}a' = -S_T,$$

$$S_T = -12\cos x - 8\cos 2x + \frac{24\sin 2x}{x} - \frac{48\sin^2 x}{x^2} + \frac{16}{x^2} - \frac{16\sin 2x}{x^3} + \frac{16\sin^2 x}{x^4}.$$

Regularity, $\nu, H \to 0$ as $t \to 0$, removes the homogeneous $n_2 \propto 1/t$, $a \propto 1/t$ and $a = \text{const}$; using (Ham) rather than (N) leaves no free t^2 in the lapse. All Si terms cancel. With $\mathrm{Cin}(z) = \int_0^z (1-\cos u)\mathrm{d}u/u$,

$$n_2 = -\cos 2x - 6\cos x + 6j_0 + 3\frac{\sin 2x}{x} - 5j_0^2, \tag{5}$$

$$a = -78 - 36\cos x - 6\cos 2x + 72 j_0 + 36\frac{\sin 2x}{x} - 24 j_0^2 + 24\,\mathrm{Cin}(2x). \tag{6}$$

These satisfy $a = 2x^2 + O(x^4)$, $n_2 = \frac{5}{3}x^2 + O(x^4)$, the exact gradient data. Assembling (4), the terms linear in $\cos x$ cancel:

$$\boxed{F_2^{\mathcal{R}}(q, K-q) \xrightarrow{K\to 0} \tfrac{1}{2}I(x) - \tfrac{1}{8}\Big(\mu^2 - \tfrac{1}{3}\Big)W(x), \quad \begin{array}{l} I = 4 + \cos^2 x - 5j_0^2, \\ W = 24\,\mathrm{Cin}(2x) + 6\cos^2 x - 54 + 48 j_0^2. \end{array}} \tag{7}$$

So the response is K^0. Here $W = a + \frac{1}{2}h^2$ and $S \to W\,\mathrm{TF}(\hat{q}\hat{q})$. Writing $F_2 \to \alpha + \beta P_2(\mu)$ with $\alpha = I/2$ and $\beta = -W/12$,

$$\alpha = \frac{x^2}{3},\ \ \beta = -\frac{x^2}{6} \quad (x \ll 1); \qquad \alpha \to 2 + \tfrac{1}{2}\cos^2 x,\ \ \beta \to -2\big[\ln 2x + \gamma_E - \tfrac{9}{4} + \tfrac{1}{4}\cos^2 x\big] \quad (x \gg 1). \tag{8}$$

This is valid for $K \ll q$ and $Kt \ll 1$ at fixed t, including $qt \gg 1$.

The power spectrum. The $\hat{q}$ average of P_2 vanishes, but that of $|F_2|^2$ does not. With $\langle P_2^2\rangle = \frac{1}{5}$,

$$\boxed{P_{22}(K\to 0) = A_0(t) = \frac{1}{\pi^2}\int_0^\infty q^2 \mathrm{d}q\, P_{\zeta_1}^2(q)\left[\alpha^2 + \frac{\beta^2}{5}\right] > 0.} \tag{9}$$

Early on, $A_0 \to \frac{7t^4}{540\pi^2}\int q^6 P^2 \mathrm{d}q$. Late, the bracket is dominated by $\frac{4}{5}\ln^2(qt)$. A_0 is dimensional power; the dimensionless spectrum is $K^3 A_0/2\pi^2$. The kernel depends on $\hat{K}$ but the spectrum is isotropic.

Why P_{13} is absent. $P_{13}(K) = 2\,\mathrm{Re}\langle X_1(K) X_3^*(K)\rangle$, and $X_1(K) \propto \zeta_1(K) = 0$ for $K < q_{\min}$, so it vanishes identically, as does P_{12}. This does not carry over to C_ℓ: $a_{\ell m}^{(1)}$ projects band modes onto ℓ through Bessel functions at finite distance, so $C_\ell^{(13)} \neq 0$ (main report).

Why the usual suppression fails. In $\dot{\mathcal{R}} \sim K^2 \Pi_S$, the scalar stress potential obeys $\Pi_S \sim \hat{K}\hat{K}\Pi/K^2$, so the K^2 cancels. A fixed tensor has helicity power $1:2:2$ (scalar:vector:tensor) over directions $\hat{K}$, so this is the same phenomenon as the K^3 tail of induced gravitational waves [1, 2].

Where each contribution comes from.

Primordial variable	None: $\mathcal{R} = \zeta$ at $t \to 0$, $\zeta_2 \equiv 0$. A local $\zeta \to \zeta + f\zeta^2$ would add a constant; here $f = 0$ by construction.
Constraint-required data at t_i	$F_2(x_i) = x_i^2/3 - \frac{1}{4}x_i^2(\mu^2 - \frac{1}{3}) + O(x_i^4)$. It is part of the same regular solution, vanishes as $t_i \to 0$, and is not an independent mode.
Measurement definition (threading)	Local products $h(1+\nu_1) - h^2/12 - \frac{1}{8}(\mu^2 - \frac{1}{3})h^2$ in K_2. They are bounded and contain the $\cos x$ terms that cancel against n_2 and a.
Evolution	n_2 (bounded, zero-mean oscillation at late times) and $-\frac{1}{4}(\mu^2 - \frac{1}{3})a$ (memory integral of the stress, with the logarithm).

Threading. In material threading, $\zeta + \nu$ has only n_2 at $K \to 0$, so β is threading dependent, and α can be removed by a local redefinition. $\mathcal{R}$ alone therefore does not establish a physical signal.

3 Second route: exact identities

Proposition 1 (All orders, for smooth irrotational flow)**.**

(a) Conservation. $\partial_t \det\gamma = 0$, *so* $\zeta_K(t) = \zeta_K(0)$ *for all* K. *Physically,* $e^{3\zeta}$ *is the entropy per unit label. Below the band* $\zeta_K = 0$ *exactly, with no singular term, no constant and no power of* K.

(b) Relation to $\mathcal{R}$. *Eq.* (3). *Thus (a) does not constrain* $\mathcal{R} - \zeta$: ν, *the composition with* X, *and* $\ln J$ *all contribute at* K.

(c) Transversality. *Define* $\mathcal{S}$ *by* $\ln(J^{-2/3}(\partial X)^T e^{H\circ X}\partial X) = \mathcal{S} + 2\,\mathrm{sym}\,\partial\xi - \frac{2}{3}\delta\,\partial\cdot\xi$. *Then* $\partial_i\partial_j\chi_{ij} = 0$ *gives, for* $K \neq 0$, $(\partial\cdot\xi)_K = -\frac{3}{4}\hat{K}_i\hat{K}_j\mathcal{S}_{ij,K}$. *Therefore*

$$\mathcal{R}_K = \mathcal{C}_K - \tfrac{1}{4}\hat{K}_i\hat{K}_j\mathcal{S}_{ij,K}, \qquad \mathcal{C} = \zeta\circ X + \nu\circ X + \tfrac{1}{3}(\ln J - \partial\cdot\xi). \tag{10}$$

(d) Shear balance, retaining all short gradients. *With the exact shear* $\Sigma^i{}_j = \frac{1}{2}\gamma^{ik}\partial_t\gamma_{kj}$ *(not to be confused with the second-order tensor* S*), and since* $\partial_t\sqrt{\gamma} = 0$, *equation (T) gives*

$$\partial_t\big(N^2\sqrt{\gamma}\,\Sigma^i{}_j\big) = -N^2\sqrt{\gamma}\ \mathrm{TF}\left[R^i{}_j[\gamma] - 2D^iD_j\nu + 2D^i\nu D_j\nu\right]. \tag{11}$$

The right side has a nonzero long-wavelength component made from short modes: this is the effective stress.

(e) Relation to the observable. *The exact redshift (main report) is* $T_{\mathrm{obs}} = T_{\mathrm{fluid}}(o)\,e^{-I}$, $I = \int\sigma_{ij}e^ie^j\mathrm{d}t$. *With* $\hat{e} = e^\zeta e$, *the integrand is* $\frac{1}{2}(\partial_t e^H)_{ij}\hat{e}^i\hat{e}^j$, *where* $\hat{e}^T e^H\hat{e} = 1$. *By (a) it has no trace part, and* ζ *does not appear in it explicitly. So* ζ *reaches* Θ *only through the ray path and as a source of* H *and* ν, *and* ν *only through the emission and observation endpoints.*

What this proves, and what it does not. Only (a) is an all-orders constraint on the response, and it concerns ζ. Part (c) is algebra on the chart definition and (d) is equation (T), so they locate the constant without fixing it. If $\mathcal{S}$ and $\mathcal{C}$ have bounded spectra at small K, (10) excludes a singular enhancement. It does *not* exclude a constant; it identifies the constant as the spectrum of $\mathcal{C}$ plus the $\hat{K}\hat{K}$ projection of $\mathcal{S}$. Stronger suppression is false, because the $O(\epsilon^2)$ value of that constant is the nonzero (7), with $\mathcal{S}_2 = W\,\mathrm{TF}(\hat{q}\hat{q})$, the same W. The boundedness hypothesis is checked only at $O(\epsilon^4)$; beyond that, (c) makes $\xi_K \sim 1/K$ at second order, which re-enters $\mathcal{S}$, and the resulting convolutions are not computed. Parts (a), (b), (d), (e) are exact for any smooth solution; (c) also needs the transverse chart to exist. No gradient expansion of the short modes and no soft input is used.

An independent value for the logarithm. The $K \to 0$ mode of (d) needs only first-order fields, and at $O(\epsilon^2)$ it is the equation for a above. The late-time mean of S_T is $-8/x^2$. Identifying $a'' + 2a'/t = (6/t^2)\Pi/\bar{\rho}$ gives $\Pi_{ij}/\bar{\rho} = 4\,\mathrm{TF}(\hat{q}\hat{q})$. This is the acoustic stress $\frac{4}{3}\bar{\rho}\langle v_i v_j\rangle$ with $v = \sqrt{3}\sin x\,\hat{q}\,\zeta_1$, and it yields $a \simeq 24\ln x$. So the $-8 = -\frac{1}{3}\cdot 24$ in $\mathcal{Q}_2$ is fixed independently of the closed forms; the bounded terms are not.

Order of limits and nonuniformity. The order is: the $O(\epsilon^2)$ kernel; then $K \to 0$ at fixed t; then, optionally, $qt \to \infty$. Exchanging $K \to 0$ and $\epsilon \to 0$ would need higher-order spectra bounded uniformly as $K \to 0$, and is an assumption. For Θ each order is finite (Prop. 2), though uniformity in ϵ is not shown; for $\mathcal{R}$ even that is the unchecked hypothesis above. Two further nonuniformities: (i) Fixed K with $t \to \infty$ leaves the regime $Kt \ll 1$, and the logarithm then saturates into propagating modes, which this calculation does not treat. (ii) Fixed ϵ with $qt \to \infty$ fails through acoustic steepening; a simple-wave estimate gives $\epsilon x_s \approx \frac{1}{2}$ for $\zeta = 2\epsilon\cos qy$. After a shock, (a) fails with entropy production. The results therefore need $\epsilon qt \ll 1$.

4 The measured temperature

By Prop. 1(e), $T_{\mathrm{fluid}}(o)$ cancels on normalisation and only shear produces anisotropy, so ν_2 and α enter only at higher order or not at all. The observer is the fluid line at fixed label X_o, with fixed proper time after $T = T_I$, and Θ is the normalised map, with the dipole removed at the end. Fields are Fourier analysed in X_o, since the observer is a label. In the chart x an extra bounded term $\xi_1 \cdot \partial\Theta_1$ would appear.

Proposition 2 (Locality fixes the form, not the coefficient)**.** *Order by order in ϵ, for compact spectra, the kernels of ν_r, H_r and Θ are entire in the Cartesian momenta, and $\Theta(\hat{n}, X_o)$ depends only on the past light cone of the observer. Hence every $P_\Theta^{(r,s)}(K)$ is continuous at $K = 0$: there is no pole, and a constant is the generic outcome.*

Sketch. First-order kernels are entire in q, since h is even with $h(0) = 0$. The induction of the main report's Cartesian proposition extends this to ν_r, H_r. The map consists of finite time integrals of these with phases $e^{ik\cdot\hat{n}\lambda}$, plus polynomial root, clock and normalisation terms. Wick contractions then integrate over compact sets. This does not apply to $\mathcal{R}$, which has the $\hat{K}\hat{K}$ projector and ξ.

So a K^0 spectrum in Θ does not by itself show a generated long mode; the squared Doppler term has one too. Also, the $K \to 0$ limit of the temperature kernel is its value at $K = 0$, with no dependence on $\hat{K}$. The pair $(q, -q)$ is plane symmetric, and the second-order map is a function of $\mu = \hat{n}\cdot\hat{q}$ alone, even in μ, with no dipole. For the seed,

$$\Theta(\hat{n}, X_o) \supset \frac{AB}{2}\cos(K\cdot X_o)\sum_{\ell\geq 2\ \mathrm{even}} \mathcal{Q}_\ell\, P_\ell(\hat{n}\cdot\hat{q}). \tag{12}$$

The ray. Set $q = 1$, $\lambda = t_o - t$, $w^2 = 1 - \mu^2$, $v = \frac{1}{2}h'$, and label the modes by $s = \pm$, with $\zeta = z_+ e^{iy} + z_- e^{-iy}$ and phase $E_s = e^{is\mu\lambda}$ along the unperturbed ray $y = \mu\lambda$. Null rays ignore N, so we may use $-\mathrm{d}t^2 + e^{2A}\mathrm{d}y^2 + e^{2B}\mathrm{d}x_\perp^2$ with $A = \zeta + f/3$, $B = \zeta - f/6$ and $H = f\,\mathrm{TF}(\hat{y}\hat{y})$. With m the local direction cosine along y, the exponent accumulated back to time t has the exact form $I(t) = \int_t^{t_o}\frac{1}{2}f'(m^2 - \frac{1}{3})\mathrm{d}t'$, and $\mathrm{d}y/\mathrm{d}t = -e^{-A}m$. Conservation of transverse momentum makes $\sqrt{1-m^2}\,e^B e^{I(t)}$ constant along the ray.

Writing $m = \mu + \delta\mu$, $y = \mu\lambda + \delta y$, for each mode

$$\delta\mu_s = w^2 \int_t^{t_o} \big(\partial_y B + \tfrac{1}{2}\mu f'\big)_s \mathrm{d}t', \qquad \delta y_s = \int_t^{t_o} \big(\delta\mu_s - \mu A_s\big)\mathrm{d}t'. \tag{13}$$

Sound waves leave only endpoints. Let $\Phi = 4j_1(x)/x$, twice the Newtonian potential, so $\Phi, \Phi' = O(x^{-2})$ are integrable. By the Bessel equation for j_0, the three integrands that drive the ray are exact derivatives up to Φ:

$$\begin{aligned} \sigma_s \equiv v\big(\mu^2 - \tfrac{1}{3}\big)E_s &= \partial_t(D_s E_s) + \Phi' E_s, \qquad D_s = \tfrac{1}{2}h'' + is\mu v, \\ \big[is(1 - \tfrac{h}{6}) + \mu v\big]E_s &= \partial_t(isvE_s) + is\Phi E_s, \\ \big[-isw^2 v - \mu(1 + \tfrac{h}{3})\big]E_s &= \partial_t(Y_s E_s) - \mu\Phi E_s, \qquad Y_s = -\tfrac{is}{2}h - \mu v. \end{aligned} \tag{14}$$

This disposes of the sound-cone resonance at $\mu^2 = \frac{1}{3}$: a total derivative cannot resonate, and the integrable Φ cannot grow. Thus, with $W_s = \int_t^{t_o} \Phi E_s \mathrm{d}t'$ and $\widetilde{W}_s = \int_t^{t_o} W_s \mathrm{d}t'$,

$$\begin{aligned} J_s &\equiv \int_{t_\star}^{t_o} \sigma_s \mathrm{d}t = \big[D_s E_s\big]_{t_\star}^{t_o} + \int_{t_\star}^{t_o} \Phi' E_s \mathrm{d}t, \\ \delta\mu_s &= isw^2(v_o - vE_s + W_s), \qquad \delta y_s = isw^2(v_o\lambda + \widetilde{W}_s) + Y_s(t_o) - Y_s E_s - \mu W_s. \end{aligned} \tag{15}$$

The first line is the first-order map, uniformly bounded in μ and in all times.

Assembly. The $z_+ z_-$ coefficient of the second-order exponent is

$$I_2 = \tfrac{1}{2}[a]_\star^o\big(\mu^2 - \tfrac{1}{3}\big) + \sum_s \int_{t_\star}^{t_o} \big(2\mu v E_s\, \delta\mu_{-s} + is\,\sigma_s\, \delta y_{-s}\big)\mathrm{d}t + I^{\mathrm{em}}.$$

Insert (15). In the local terms the phases cancel: the v^2 terms cancel between $s = \pm$, and the rest integrates to $\frac{1}{4}[h^2]_\star^o(\mu^2 - \frac{1}{3})$, completing a to $W = a + \frac{1}{2}h^2$. So the tensor in $\mathcal{R}$ emerges from the ray calculation by itself. With the normalisation and the clock, the quadrupole coefficient is exactly

$$\boxed{\mathcal{Q}_2 = -\tfrac{1}{3}\big[W\big]_{x_\star}^{x_o} + \mathcal{Q}_2^{\mathrm{em}} + \mathcal{Q}_2^{\mathrm{prod}} + \mathcal{Q}_2^{\mathrm{clk}} + \mathcal{Q}_2^{\mathrm{rem}},} \tag{16}$$

where P_2 projects onto the coefficient of $P_2(\mu)$ and P_0 removes the sky mean.

- *Generated:* $-\frac{1}{2}\hat{n}^i\hat{n}^j[S_{ij}]_\star^o$, with $S = W\,\mathrm{TF}(\hat{q}\hat{q})$.
- *Emission surface:* $\mathcal{Q}_2^{\mathrm{em}} = -\frac{2}{3}t_\star h'(t_\star)\nu_1(t_\star) \to 2x_\star \sin 2x_\star$ for $x_\star \gg 1$. First-order displacement of the surface times the local shear: a direct short-mode term.
- *Normalised products:* $\mathcal{Q}_2^{\mathrm{prod}} = \mathsf{P}_2\mathsf{P}_0(J_+J_-)$, with the sky means of $J_\pm$ subtracted. By (15), $|D_s|^2$ gives $\to 2(\sin^2 x_o + \sin^2 x_\star)$ for large separation: the squared Doppler terms at both ends.
- *Clock:* $\mathcal{Q}_2^{\mathrm{clk}} = -D\,\mathsf{P}_2(\partial_{t_o}J_+ + \partial_{t_o}J_-)$, with $D = [-t_I^2\nu_1(t_I) - \int_{t_I}^{t_o} t\nu_1 \mathrm{d}t]/t_o$.
- *Remainder:* $\mathcal{Q}_2^{\mathrm{rem}} = -\mathsf{P}_2 R$, with the nonlocal ray terms

$$R = \sum_s \Big[-2is\mu w^2 \int vE_s(v_o + W_{-s})\mathrm{d}t + w^2 \int \sigma_s(v_o\lambda + \widetilde{W}_{-s})\mathrm{d}t + isY_{-s}(t_o)J_s - is\mu \int \sigma_s W_{-s}\mathrm{d}t\Big]. \tag{17}$$

Theorem 1 (The generated logarithm survives the complete measurement)**.** *With* $B = \mathcal{Q}_2^{\mathrm{prod}} + \mathcal{Q}_2^{\mathrm{clk}} + \mathcal{Q}_2^{\mathrm{rem}}$ *there is an absolute constant* C *with*

$$|B| \le C\big(1 + x_I^2/x_o\big) \qquad \textit{for all } x_\star < x_o,\ x_I < x_o. \tag{18}$$

Consequently, for $x_o > x_\star \geq 1$,

$$\mathcal{Q}_2 = -8\ln\frac{x_o}{x_\star} + \mathcal{Q}_2^{\rm em} + O\big(1 + x_I^2/x_o\big), \tag{19}$$

and at fixed $x_\star, x_I$ the complete quadrupole diverges logarithmically as $x_o \to \infty$, at the rate of the generated term.

Proof. All of v, h'', D_s, Y_s are bounded, and $\|\Phi\|_1, \|\Phi'\|_1 < \infty$ on $(0,\infty)$. So $|J_s| \leq C$ uniformly by (15), which bounds $\mathcal{Q}_2^{\rm prod}$. Likewise $\partial_{t_o} J_s = \sigma_s(t_o) + is\mu J_s$ is bounded. Since $\int t\cos x\,\mathrm{d}t = O(t_o) + O(t_I)$ and tj_0 integrates to a bounded function, $|D| \leq C(1 + t_I^2/t_o)$, which bounds $\mathcal{Q}_2^{\rm clk}$.

For $\mathcal{Q}_2^{\rm rem}$ we need one lemma. For a polynomial p, $K_p(\tau) = \int_{-1}^{1} p(\mu)w^2 P_2(\mu)e^{i\mu\tau}\mathrm{d}\mu = O((1+|\tau|)^{-2})$: integrate by parts twice, the first boundary term vanishing because $w^2 = 0$ at $\mu = \pm 1$. Now take the four terms of (17), after P_2.

(i) The first term carries w^2 and the phase $E_s(t)$, which becomes $e^{is\mu(t'-t)}$ inside W_{-s}. Its part in v_o is at most $\|v\|_\infty\|K_p\|_1$ and its part in W_{-s} at most $\|v\|_\infty\|K_p\|_1\|\Phi\|_1$. (ii) In the second term, use (14) to write $\int \lambda\sigma_s\mathrm{d}t = -\lambda_\star D_s E_s|_\star + \int D_s E_s\mathrm{d}t + \int \lambda\Phi' E_s\mathrm{d}t$, where $\lambda_\star = t_o - t_\star$. These are bounded by $\sup_\tau \tau|K_p(\tau)|$, $\|D_s\|_\infty\|K_p\|_1$ and $\|\Phi'\|_1 \sup_\tau \tau|K_p(\tau)|$. The part in $\widetilde{W}_{-s} = \int_t^{t_o}(t''-t)\Phi E_{-s}\mathrm{d}t''$ has $\tau = t''-t$ and $\tau|K_p(\tau)| \leq C/(1+\tau)$, so it is at most $C\int \mathrm{d}t''|\Phi(t'')|\ln(1+t'') < \infty$, since $\Phi = O(t^{-2})$. (iii) The third term is bounded because Y_s and J_s are. (iv) In the fourth, exchange the integrals: $\int \sigma_s W_{-s}\mathrm{d}t = \int \mathrm{d}t'\,\Phi(t')E_{-s}(t')\int_{t_\star}^{t'}\sigma_s\mathrm{d}t$, and the inner integral is uniformly bounded, by the first line of (15), so this is at most $C\|\Phi\|_1$.

For (19), $W(x) = 24\ln x + O(1)$ for $x \geq 1$. □

The visible oscillation in the $O(1)$ is $4\sin^2 x_o$: $2\sin^2 x_o$ from the $\cos^2 x$ in W and $2\sin^2 x_o$ from $\mathcal{Q}_2^{\rm prod}$, the second-order boost of the observer.

Outside the horizon. For $qt \ll 1$ everything is local, and

$$\mathcal{Q}_2 = -\tfrac{2}{9}q^2(t_o^2 - t_\star^2) + O(q^4t^4), \tag{20}$$

which is the $z_1 z_2$ part of the local quadrupole $-\frac{1}{6}\Delta(t^2)\,e^{-2\zeta}\,\mathrm{TF}[\partial\partial\zeta - \partial\zeta\partial\zeta]_{nn}$. It is a nonzero K^0 response, not suppressed relative to K^2 the way a linear long adiabatic mode is.

The spatial spectrum of the temperature. Writing the $\ell = 2$ part as $Q_{ij}\hat{n}^i\hat{n}^j$,

$$P_{Q_{ij}Q_{ij}}(K \to 0) = \frac{3\epsilon^4}{8\pi^2}\int q^2\mathrm{d}q\,P_{\zeta_1}^2(q)\,\mathcal{Q}_2^2(qt_\star, qt_o, qt_I) > 0. \tag{21}$$

The inequality holds for every nontrivial band, including subhorizon ones: by Prop. 2, $\mathcal{Q}_2$ is real analytic in q, the small-q form above shows that it is not identically zero, and so it cannot vanish on an interval. This is white noise in the observer-label field of the measured quadrupole, not an angular spectrum; C_2 receives this region only with phase-space weight $\propto \int K^2\mathrm{d}K$.

The power depends on $\mathcal{Q}_2^2$, so the zero mean of the emission term gives no protection. By Theorem 1, with $r = x_o/x_\star$ and $x_I^2 \lesssim x_o$, the band average has $(8\ln r)^2$ from the generated term and $\langle(2x_\star\sin 2x_\star)^2\rangle \simeq 2x_\star^2$ from emission, up to bounded corrections. So the generated term dominates for $x_\star \ll \ln r$ and the emission term for $x_\star \gg \ln r$. Analyticity shows white noise in the total quadrupole; only the theorem identifies its generated part.

What is and is not curvature here. A constant-in-time long S_{ij} is a coordinate rescaling, invisible in Θ, which sees only ΔS. Removing that part does not make the connected spectrum vanish, because the stress makes S_{ij} time dependent. The adiabatic expectation $\Theta_2 \sim K^2t^2\mathcal{R}$ fails because this $\mathcal{R}$ arrives with an order-K^0 shear. $\mathcal{R}$ and Θ read the same tensor, via $\hat{K}\hat{K}$ and $\hat{n}\hat{n}$.

A Numerical corroboration

No argument above depends on these checks.

Spatial limit. The exact kernels at finite K converge to (7) at all tested angles, including $|q_1| \neq |q_2|$. A direct quadrature for a smooth compact bump centred at q_0 with half-width $0.3q_0$ gives $P_{22}(K)/A_0 = 0.8589, 0.9565, 0.9867$ at $K/q_0 = 0.2, 0.1, 0.05$ for $q_0 t/\sqrt{3} = 1$, and $0.7685, 0.9327, 0.9782$ for $q_0 t/\sqrt{3} = 4$. The deficit shrinks by a factor of about 3.3 per halving of K; we do not claim the exact power of the correction. On the shell, the $K \to 0$ limit of the Kohri–Terada form in the main report satisfies $F_T \to a/2$, agreeing to 3×10^{-11} for x up to 15. Formula (4) is verified off shell, for generic momenta and an arbitrary trace-free H_2, by matching the scalar curvature in the two charts.

The theorem's ingredients. Identities (14) and $W = a + \frac{1}{2}h^2$ are checked by exact computer algebra. The printed R of (17) reproduces the original perturbative ray map, with differences at most about 10^{-5}. Table 1 gives the evaluated pieces; B there stays small, $+3.1$ at $(5, 40)$ for example.

$x_\star$	x_o	$\mathcal{Q}_2$	$-\frac{1}{3}[W]$	rem	em	prod	clk
0.1	0.2	−0.019594	−0.0201	+0.0003	−0.0000	+0.0000	+0.0002
0.5	1	−0.319767	−0.5230	+0.1519	−0.0251	+0.0122	+0.0642
1	3	−4.268883	−2.7998	+1.3472	−0.2881	+0.1195	−2.6478
3	10	+0.292937	−8.1918	+0.1877	+6.8473	+0.8380	+0.6116
5	20	−15.3640	−10.1915	−1.8555	−7.3100	+3.4488	+0.5442
5	40	−20.7210	−16.5168	+1.7639	−7.3100	+1.5628	−0.2209
5	80	−24.8166	−21.0832	+0.3781	−7.3100	+3.0030	+0.1955
5	160	−33.9877	−28.5273	+0.9593	−7.3100	+0.9047	−0.0144

Table 1: The $K \to 0$ quadrupole coefficient in (16), $t_I = t_\star/2$, by quadrature.

Forward test, with no perturbative map. We solved the exact plane-symmetric Einstein–radiation system with $\zeta = 2\epsilon \cos qy$, integrated exact null rays with physical emission roots, the proper-time clock and the exact sky normalisation, and averaged over observer labels. Richardson extrapolation in ϵ reproduces $\mathcal{Q}_2$ with differences 3×10^{-8} to 5×10^{-6} for seven time pairs up to $(x_\star, x_o) = (5, 20)$, and at most 8×10^{-5} for $(5, 40), (2, 40), (5, 80), (2, 80), (5, 160)$. Omitting any single piece of the map is detected.

The same solver tests $K \neq 0$, with $\zeta = 2\epsilon[\cos(ny) + \cos((n-1)y)]$ in units $K = 1$: our seed with $\hat{q} \parallel \hat{K}$, read from the observer-label harmonic $\cos y_0$. With times set by the mean q, $\mathcal{Q}_2(K) - \mathcal{Q}_2$ falls as K^2 (Table 2). Other angles are not tested forward. At small K the initial data must be accurate in the K channel, because (N) has the constraint-violating homogeneous solution $\nu \propto t^2$. With gradient data alone, an $O(x_s^4)$ error at the start time x_s grew into a spurious $K \to 0$ offset of about -0.44 at $(3, 10)$. An exact series through t^8 removes it.

K/q (n)	1/7.5 (8)	1/15.5 (16)	1/31.5 (32)	1/63.5 (64)
$(x_\star, x_o) = (1, 3)$	+0.294	+0.0697	+0.0169	+0.0042
$(3, 10)$	–	+0.249	+0.0672	+0.0170
$(5, 20)$	–	–	+2.77	+0.706

Table 2: Forward test at $K \neq 0$: $\mathcal{Q}_2(K) - \mathcal{Q}_2$ for the collinear pair, Richardson extrapolated in ϵ.

Late times. Fitting $\mathcal{Q}_2 + \frac{1}{3}[W] - \mathcal{Q}_2^{\mathrm{em}}$ over log-spaced $x_o \in (30, 1000)$ gives a logarithm coefficient of -0.03 ± 0.02, 0.00 ± 0.12, $+0.09 \pm 0.11$ for $x_\star = 2, 5, 8$ (bootstrap standard deviations of the fit). This is finite-range evidence, consistent with Theorem 1 but not a bound on the true asymptotic coefficient. A regression of $\mathcal{Q}_2$ itself gives a $\cos 2x_o$ coefficient of $-2.01, -2.03, -1.79$, against the -2 from $4 \sin^2 x_o$.

Division of the power. Table 3 evaluates the competition at finite $x_\star$.

$x_\star$	$x_o/x_\star$	rms $\mathcal{Q}_2$	rms gen	rms em	$\langle \mathrm{gen}^2\rangle/\langle\mathcal{Q}_2^2\rangle$	$\langle \mathrm{em}^2\rangle/\langle\mathcal{Q}_2^2\rangle$
1	4	3.12	3.66	0.34	1.38	0.01
3	4	4.49	9.80	6.82	4.77	2.31
5	4	10.60	10.89	6.86	1.06	0.42
10	4	16.08	11.06	14.12	0.47	0.77
20	4	29.81	11.12	29.22	0.14	0.96
5	32	24.83	27.43	6.86	1.22	0.08

Table 3: Division of the temperature white-noise power, with q averaged over $\pm 30\%$ with flat weight at fixed $t_o/t_\star$ and $t_I = t_\star/2$. Fractions need not sum to one because of cross terms.

Steepening. Exact plane-symmetric evolution agrees with the simple-wave estimate: the $K = 0$ average satisfies $\langle f\rangle/\epsilon^2 - a \approx 0.39$ ($\epsilon = 0.01$, $x = 10$), 3.39 ($\epsilon = 0.03$, $x = 10$), scaling as ϵ^2, and spectral steepening appears at $\epsilon x \approx 0.3$–0.4.

B Status of the claims

Claim	Status
ζ conservation, identities (a)–(e)	Exact derivation
Theorem 1 (boundedness of B, surviving logarithm)	Proved by hand; identities (14) by exact computer algebra; (17) compared with the map
$P_{Q_{ij}Q_{ij}} > 0$ for every band	Proved, given Prop. 2
a, n_2, I, W, $F_2^{\mathcal{R}}$ limit, (4)	Exact symbolic check, negative controls
$K \to 0$ continuity of field kernels	Proved (entire Green function); numerically confirmed
$\mathcal{Q}_2$ exact formula (16), (17)	Derived; forward test at $K = 0$ up to $x_o = 160$ (corroboration)
$\mathcal{Q}_2(K) \to \mathcal{Q}_2$ at $K \neq 0$, $\hat{q} \parallel \hat{K}$	Forward test; K^2 convergence
Finite $K \to 0$ limit of Θ at each order (Prop. 2)	Proof sketch; forward check at second order
Exchange of $K \to 0$ and $\epsilon \to 0$; bounded $\mathcal{R}$ spectrum beyond $O(\epsilon^4)$	Assumed, not shown
Long-time behaviour beyond $\epsilon q t \ll 1$; shocks	Not resolved; outside the smooth finite-time setup of the question
Forward test with $\hat{q}$ at a generic angle to $\hat{K}$	Not performed (needs 3+1 evolution); $\mathcal{R}$ kernels at generic angle are checked by the Green solution
Complete C_ℓ from this white noise; superhorizon-to-subhorizon entry of the output	Not computed; the question does not require them

C Code and reproduction

All code is in `solution/followup/`; `run_followup.sh` runs everything (about 40 minutes, peak memory about 350 MB).

File	Purpose
`plane_reduction.py`, `k0_fields.py`	reduced equations from the Ricci formula; closed-form proofs
`check_chart_R.py`	off-shell scalar-curvature invariance of (4)
`check_followup.py`, `check_k0_odes.py`	negative controls, tensor limit, stress, helicity, omission tests; the $K = 0$ source equations
`check_temperature_theorem.py`	identities (14), $W = a + \frac{1}{2}h^2$, printed R against the map
`map_k0.py`	complete $\mathcal{Q}_2$ and pieces, Table 1
`forward_plane.py`, `forward_log.py`	exact PDE and exact rays against $\mathcal{Q}_2$, including the logarithmic regime
`forward_K.py`, `k_series.py`	the same at $K \neq 0$; exact t-series of the K-channel initial data
`finite_K.py`	finite-K limit and $P_{22}(K)/A_0$
`late_time.py`, `late_log_bound.py`	regression for (19); bound on a further logarithm
`power_split.py`, `nonuniform.py`	Table 3; amplitude nonuniformity

The forward solver takes f' at nonzero wavenumber from the momentum constraint: evolving (T) directly excites the $-K^2/3$ scalar instability excluded by the constraints. At small K the initial data must also be accurate in the K channel, because (N) has the constraint-violating homogeneous solution $\nu \propto t^2$. With gradient data alone, an $O(x_s^4)$ error at the start time x_s grew into a spurious $K \to 0$ offset of about -0.44 at $(3, 10)$. The series through t^8 removes it.

A stand-alone check of the decisive fields (`minimal_check.py`), which needs only SymPy and prints zero residuals:

```
"""Stand-alone: the K->0 pair fields solve the plane-symmetric equations (q=1, x=t/sqrt3)."""

# fmt: off
import sympy as sp

t, y, e, z1, z2 = sp.symbols("t y e z1 z2", positive=True)
x = t / sp.sqrt(3)
j0, c = sp.sin(x) / x, sp.cos(x)
C2 = sp.Function("C2")(t)                       # Cin(2x); C2'(t) = (1-cos 2x)/t
n1, h1 = c - j0, 6 * (1 + c - 2 * j0)
n2 = -sp.cos(2 * x) - 6 * c + 6 * j0 + 3 * sp.sin(2 * x) / x - 5 * j0**2
a2 = -78 - 36 * c - 6 * sp.cos(2 * x) + 72 * j0 + 36 * sp.sin(2 * x) / x - 24 * j0**2 + 24 * C2
g = z1 * sp.exp(sp.I * y) + z2 * sp.exp(-sp.I * y)
Z = e * g
f = e * h1 * g + e**2 * a2 * z1 * z2             # H = f diag(2/3,-1/3,-1/3); gamma = e^{2Z} exp H
nu = e * n1 * g + e**2 * n2 * z1 * z2
D = lambda u, v: sp.diff(u, v).subs(sp.Derivative(C2, t), (1 - sp.cos(2 * x)) / t)
fy, ny, zy = D(f, y), D(nu, y), D(Z, y)
ex = sp.exp(-2 * f / 3 - 2 * Z)
Ff = ex * (-fy**2 / 18 + fy * ny / 3 + fy * zy / 6 + D(fy, y) / 6 + 2 * ny**2
           + 4 * ny * zy - 2 * D(ny, y) + zy**2 - D(zy, y))
ft, nt = D(f, t), D(nu, t)
lap = ex * (D(ny, y) - (zy + fy / 3) * ny + 2 * (zy - fy / 6) * ny)
T_ = D(ft, t) + 2 * (1 / t + nt) * ft + 2 * Ff                                  # (T)
N_ = (3 * D(nt, t) - lap - 2 * ex * ny**2 + ft**2 / 6
      + 3 / t**2 * (sp.exp(-2 * nu) - 1))                                       # (N)
for name, eq in (("T", T_), ("N", N_)):
    first = sp.simplify(sp.diff(eq, e).subs(e, 0))
    second = sp.expand(sp.diff(eq, e, 2).subs(e, 0) / 2).coeff(z1, 1).coeff(z2, 1)
    print(name, "first:", first, "pair:", sp.simplify(sp.expand(sp.expand_trig(second))))
```

References

[1] K. Kohri and T. Terada, Phys. Rev. D 97 (2018) 123532.

[2] R.-G. Cai, S. Pi and M. Sasaki, Phys. Rev. D 102 (2020) 083528.